AI.101

Introduction to Artificial Int

COURSE OVERVIEW

Origins, definitions, and the landscape of artificial intelligence; how AI perceives, reasons, learns, and acts, and its role in modern engineering.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: WHAT IS AI?

KEY TOPICS

- Defining AI: narrow vs. general vs. super
- A brief history of AI and robotics
- AI in everyday systems
- Course scope and learning path

TEACHING FOCUS

Situate AI within engineering and society; set expectations for the 6-hour orientation.

KEY TAKEAWAY AI = machines that perceive, reason, learn, act.

2

SESSION

60 MINUTES

INTELLIGENT AGENTS & ENVIRONMENTS

KEY TOPICS

- Agent architecture: sensors, actuators, effectors
- Rational agents and performance measures
- Environment types: observable, deterministic, episodic
- PEAS framework in robotics

TEACHING FOCUS

Model an intelligent agent and its operating environment.

KEY TAKEAWAY Every intelligent system is an agent in an environment

3

SESSION

60 MINUTES

PROBLEM SOLVING & SEARCH

KEY TOPICS

- Uninformed search: BFS, DFS, UCS
- Informed search: greedy, A*, heuristics
- Adversarial search: minimax, alpha-beta
- Constraint satisfaction intro

TEACHING FOCUS

Apply classical search to robot path and puzzle problems.

KEY TAKEAWAY Search is the engine of automated reasoning.

4

SESSION

60 MINUTES

KNOWLEDGE REPRESENTATION & LOGIC

KEY TOPICS

- Propositional and first-order logic
- Semantic networks and frames
- Ontologies for robotics domains
- Reasoning under uncertainty

TEACHING FOCUS

Encode domain knowledge so machines can infer.

KEY TAKEAWAY Knowledge plus inference equals intelligence.

5

SESSION

60 MINUTES

PLANNING & DECISION MAKING

KEY TOPICS

- STRIPS and classical planning
- State-space vs. plan-space planning
- Decision trees and utility
- Introduction to Markov decision processes

TEACHING FOCUS

Generate action sequences to achieve robot goals.

KEY TAKEAWAY Planning turns goals into action sequences.

6

SESSION

60 MINUTES

AI APPLICATIONS, ETHICS & REVIEW

KEY TOPICS

- AI in industry, healthcare, transport
- Bias, transparency, and accountability
- Course recap and assessment
- Next steps in the degree

TEACHING FOCUS

Connect foundations to real-world impact and responsibility.

▲ **KEY TAKEAWAY** Intelligent systems must be built responsibly.